Genetics and Evolution

The work of Mendel and others who followed him gave us an idea of inheritance patterns, but the nature of those factors which determine the phenotype was not very clear.

What became the focus

- as these factors represent the genetic basis of inheritance, understanding the structure of the genetic material, and the structural basis of genotype and phenotype conversion, became the focus of attention in biology for the next century
- the entire body of molecular biology was a consequent development, with major contributions from Watson, Crick, Nirenberg, Khorana, the Kornbergs father and son, Benzer, Monod and Brenner

The parallel problem

- a parallel problem being tackled was the mechanism of evolution
- awareness in the areas of molecular genetics, structural biology and bio informatics has enriched our understanding of the molecular basis of evolution
- in this unit the structure and function of DNA, and the story and theory of evolution, have been examined and explained

In this unit

Ch 4 Principles of Inheritance and Variation

Ch 5 Molecular Basis of Inheritance

Ch 6 Evolution

James Dewey Watson

- born in Chicago on 6 April 1928, he received a B.Sc. degree in Zoology in 1947, during which years his interest in bird-watching had matured into a serious desire to learn genetics
- this became possible when he received a Fellowship for graduate study in Zoology at Indiana University, Bloomington, where he received his Ph.D. degree in 1950 on a study of the effect of hard X-rays on bacteriophage multiplication
- he met Crick and discovered their common interest in solving the DNA structure, and their first serious effort was unsatisfactory, while their second, based on more experimental evidence and a better appreciation of the nucleic acid literature, resulted early in March 1953 in the proposal of the complementary double-helical configuration

Francis Harry Compton Crick

- born on 8 June 1916 at Northampton, England, he studied physics at University College, London and obtained a B.Sc. in 1937
- he completed a Ph.D. in 1954 on a thesis entitled X-ray Diffraction: Polypeptides and Proteins
- a critical influence in his career was his friendship with J. D. Watson, then a young man of 23, leading in 1953 to the proposal of the double-helical structure for DNA and the replication scheme
- Crick was made an F.R.S. in 1959

The honours

- the John Collins Warren Prize of the Massachusetts General Hospital, in 1959
- the Lasker Award, in 1960
- the Research Corporation Prize, in 1962
- and above all, the Nobel Prize in 1962

The questions genetics asks

- why an elephant always gives birth only to a baby elephant and not some other animal, and why a mango seed forms only a mango plant and not any other plant
- whether the offspring are identical to their parents, or show differences in some of their characteristics
- why siblings sometimes look so similar to each other, and sometimes so different

The two terms

- these and several related questions are dealt with scientifically in a branch of biology known as genetics, which deals with the inheritance as well as the variation of characters from parents to offspring
- inheritance is the process by which characters are passed on from parent to progeny, and it is the basis of heredity
- variation is the degree by which progeny differ from their parents

What people knew early

- humans knew from as early as 8000-1000 B.C. that one of the causes of variation was hidden in sexual reproduction
- they exploited the variations naturally present in wild populations of plants and animals, to selectively breed and select for organisms that possessed desirable characters

Stem height, tall against dwarf

Flower colour and flower position

Pod shape and pod colour

Seed shape and seed colour

Seven pairs in all, each with two opposing traits

Figure 4.A Seven pairs of contrasting traits in pea plant studied by Mendel

Breeding before science

- through artificial selection and domestication from ancestral wild cows we have well-known Indian breeds, such as the Sahiwal cows in Punjab
- though our ancestors knew about the inheritance of characters and variation, they had very little idea about the scientific basis of these phenomena

4.1 Mendel's laws of inheritance

Why his work carried weight

- it was during the mid-nineteenth century that headway was made in the understanding of inheritance, when Gregor Mendel conducted hybridisation experiments on garden peas for seven years, from 1856 to 1863, and proposed the laws of inheritance in living organisms
- during his investigations it was for the first time that statistical analysis and mathematical logic were applied to problems in biology, and his experiments had a large sampling size, which gave greater credibility to the data he collected
- the confirmation of his inferences from experiments on successive generations of his test plants proved that his results pointed to general rules of inheritance rather than being unsubstantiated ideas

True-breeding lines

- Mendel investigated characters in the garden pea plant that were manifested as two opposing traits, such as tall or dwarf plants and yellow or green seeds, which allowed him to set up a basic framework of rules governing inheritance, expanded on by later scientists to account for all the diverse natural observations and the complexity inherent in them
- he conducted artificial pollination and cross pollination experiments using several true-breeding pea lines, a true-breeding line being one that, having undergone continuous self-pollination, shows stable trait inheritance and expression for several generations
- he selected 14 true-breeding pea plant varieties as pairs which were similar except for one character with contrasting traits, some of the contrasting traits selected being smooth or wrinkled seeds, yellow or green seeds, inflated or constricted green or yellow pods, and tall or dwarf plants

Removing the anthers of the female parent before they dehisce

Bagging the emasculated flower

Dusting pollen from the chosen male parent on the stigma

Rebagging, and collecting the seed that forms

Figure 4.B Steps in making a cross in pea

4.2 Inheritance of one gene

The F1 and the F2

- Mendel crossed tall and dwarf pea plants to study the inheritance of one gene, collected the seeds produced and grew them to generate plants of the first hybrid generation, called the Filial1 progeny or F1
- he observed that all the F1 progeny plants were tall, like one of the parents, and none were dwarf, and he made similar observations for the other pairs of traits, finding that the F1 always resembled either one of the parents and that the trait of the other parent was not seen in them
- he then self-pollinated the tall F1 plants and to his surprise found that in the Filial2 generation some of the offspring were dwarf, the character not seen in the F1 now being expressed, with 1/4th of the F2 plants dwarf and 3/4th tall
- the tall and dwarf traits were identical to their parental type and showed no blending, all the offspring being either tall or dwarf and none of in-between height, and similar results were obtained with the other traits, only one parental trait being expressed in the F1 while at F2 both were expressed in the proportion 3:1

Table 4.A Contrasting Traits Studied by Mendel in Pea

S.No.	Characters	Contrasting Traits
1.	Stem height	Tall/dwarf
2.	Flower colour	Violet/white
3.	Flower position	Axial/terminal
4.	Pod shape	Inflated/constricted
5.	Pod colour	Green/yellow
6.	Seed shape	Round/wrinkled
7.	Seed colour	Yellow/green

Tall parent crossed with dwarf parent

F1 generation, all tall

F1 self-pollinated

F2 generation, three tall to one dwarf

Figure 4.C Diagrammatic representation of monohybrid cross

Factors, now called genes

- Mendel proposed that something was being stably passed down unchanged from parent to offspring through the gametes over successive generations, and he called these things factors, which we now call genes
- genes are the units of inheritance and contain the information required to express a particular trait in an organism
- genes which code for a pair of contrasting traits are known as alleles, that is, they are slightly different forms of the same gene

Symbols, genotype and phenotype

- if alphabetical symbols are used for each gene, the capital letter is used for the trait expressed at the F1 stage and the small letter for the other trait, so for height T is used for tall and t for dwarf, T and t being alleles of each other, and the pair of alleles for height would be TT, Tt or tt
- in a true breeding tall or dwarf pea variety the allelic pair of genes for height are identical or homozygous, TT and tt respectively
- TT and tt are called the genotype of the plant while the descriptive terms tall and dwarf are the phenotype, which leaves the phenotype of a plant of genotype Tt to be worked out

Dominant and recessive

- as Mendel found the phenotype of the F1 heterozygote Tt to be exactly like the TT parent in appearance, he proposed that in a pair of dissimilar factors one dominates the other and is hence called the dominant factor, while the other factor is recessive
- in this case T for tallness is dominant over t for dwarfness, which is recessive, and he observed identical behaviour for all the other character and trait-pairs he studied
- it is convenient and logical to use the capital and lower case of one alphabetical symbol to remember this concept, rather than T for tall and d for dwarf, which makes it difficult to remember whether the two are alleles of the same gene
- alleles can be similar, as in the homozygotes TT and tt, or dissimilar, as in the heterozygote Tt

Gametes of the two parents written on the two sides

All possible combinations filled into the boxes below

A square output form, showing every genotype

The probability of each genotype read off directly

Figure 4.D A Punnett square used to understand a typical monohybrid cross conducted by Mendel between true-breeding tall plants and true-breeding dwarf plants

4.3 Segregation of alleles

One allele to a gamete

- since the Tt plant is heterozygous for genes controlling one character, height, it is a monohybrid, and the cross between TT and tt is a monohybrid cross
- from the observation that the recessive parental trait is expressed without any blending in the F2 generation, we can infer that when the tall and dwarf plant produce gametes by meiosis, the alleles of the parental pair separate or segregate from each other and only one allele is transmitted to a gamete
- this segregation of alleles is a random process, so there is a 50 per cent chance of a gamete containing either allele, as verified by the results of the crossings, and the gametes of the tall TT plants have the allele T while those of the dwarf tt plants have t
- during fertilisation the two alleles, T from one parent through the pollen and t from the other through the egg, unite to produce zygotes with one T and one t allele, so the hybrids have Tt and are heterozygous

4.4 The Punnett square

What it shows

- the production of gametes by the parents, the formation of the zygotes and the F1 and F2 plants can be understood from a diagram called the Punnett square, developed by the British geneticist Reginald C. Punnett
- it is a graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross, the possible gametes being written on two sides, usually the top row and left columns, and all possible combinations represented in boxes below
- the square shows the parental tall TT male and dwarf tt female plants, the gametes produced by them and the F1 Tt progeny, the symbols for female eggs and male pollen marking the two sides
- the F1 plant of genotype Tt, when self-pollinated, produces gametes of genotype T and t in equal proportion, so on fertilisation pollen grains of genotype T have a 50 per cent chance to pollinate eggs of genotype T as well as of genotype t, and pollen grains of genotype t likewise

4.5 The ratios, and the test cross

Phenotypic and genotypic ratios

- as a result of random fertilisation the resultant zygotes can be of genotype TT, Tt or tt, and from the Punnett square it is easily seen that 1/4th of the random fertilisations lead to TT, 1/2 to Tt and 1/4th to tt
- though the F1 have a genotype of Tt the phenotypic character seen is tall, and at F2 3/4th of the plants are tall, some of them TT and others Tt, externally indistinguishable, so within the genopytic pair Tt only the character T, tall, is expressed
- it is due to this dominance that all the F1 are tall though the genotype is Tt, and in the F2 3/4th are tall though genotypically 1/2 are Tt and only 1/4th TT, leading to a phenotypic ratio of 3:1 but a genotypic ratio of 1:2:1
- the 1/4 : 1/2 : 1/4 ratio of TT : Tt : tt is mathematically condensable to the binomial expression (ax + by)2, with the gametes bearing T or t in equal frequency, expanded as (1/2T + 1/2t)2 = (1/2T + 1/2t) X (1/2T + 1/2t) = 1/4 TT + 1/2Tt + 1/4 tt

The test cross

- Mendel self-pollinated the F2 plants and found that dwarf F2 plants continued to generate dwarf plants in F3 and F4 generations, and concluded that the genotype of the dwarfs was homozygous tt, which leaves what he would have got from self-pollinating a tall F2 plant to be worked out
- though genotypic ratios can be calculated using mathematical probability, by simply looking at the phenotype of a dominant trait it is not possible to know the genotypic composition, so whether a tall plant from F1 or F2 has TT or Tt composition cannot be predicted
- therefore, to determine the genotype of a tall plant at F2, Mendel crossed the tall plant from F2 with a dwarf plant, and this he called a test cross
- in a typical test cross an organism showing a dominant phenotype, whose genotype is to be determined, is crossed with the recessive parent instead of self-crossing, and the progenies of such a cross can easily be analysed to predict the genotype of the test organism

Work it out

- using a Punnett square, find out the nature of the offspring of a test cross, and what ratio you get
- using the genotypes of this cross, give a general definition for a test cross

A violet flowered plant of unknown genotype

Crossed with the recessive white flowered parent

The progeny read off to give the unknown genotype

Violet colour V dominant over white colour v

Figure 4.E Diagrammatic representation of a test cross

4.6 Law of dominance

What it states

- characters are controlled by discrete units called factors
- factors occur in pairs
- in a dissimilar pair of factors one member of the pair dominates the other, which is recessive
- the law is used to explain the expression of only one of the parental characters in a monohybrid cross in the F1 and the expression of both in the F2, and it also explains the proportion of 3:1 obtained at the F2

4.7 Law of segregation

What it rests on

- this law is based on the fact that the alleles do not show any blending and that both the characters are recovered as such in the F2 generation, though one of them is not seen at the F1 stage
- though the parents contain two alleles, during gamete formation the factors or alleles of a pair segregate from each other, such that a gamete receives only one of the two factors
- a homozygous parent produces all gametes that are similar, while a heterozygous one produces two kinds of gametes, each having one allele, in equal proportion

Red flowered parent RR crossed with white flowered rr

F1 all pink, Rr, resembling neither parent

F2 in the ratio one red, two pink, one white

Genotype ratio unchanged, phenotype ratio changed

Figure 4.F Results of monohybrid cross in the plant Snapdragon, where one allele is incompletely dominant over the other allele

4.8 Incomplete dominance

The dog flower

- when experiments on peas were repeated using other traits in other plants, it was found that sometimes the F1 had a phenotype that did not resemble either of the two parents and was in between the two
- the inheritance of flower colour in the dog flower, snapdragon or Antirrhinum sp., is a good example, where a cross between true-breeding red-flowered (RR) and true-breeding white-flowered plants (rr) gave an F1 (Rr) that was pink
- when the F1 was self-pollinated the F2 resulted in the ratio 1 (RR) red : 2 (Rr) pink : 1 (rr) white, so the genotype ratios were exactly as expected in any mendelian monohybrid cross but the phenotype ratios had changed from the 3:1 dominant to recessive ratio
- what happened was that R was not completely dominant over r, and this made it possible to distinguish Rr as pink from RR red and rr white

4.9 What dominance actually is

Two copies of every gene

- every gene contains the information to express a particular trait, and in a diploid organism there are two copies of each gene, that is a pair of alleles
- these two alleles need not always be identical, as in a heterozygote, one of them possibly differing due to some change it has undergone which modifies the information that allele contains
- taking a gene that contains the information for producing an enzyme, and assuming as is more common that the normal allele produces the normal enzyme needed for the transformation of a substrate S, the modified allele could theoretically be responsible for the production of a normal or less efficient enzyme, or of a non-functional enzyme, or of no enzyme at all

Which allele is dominant, and why

- in the first case the modified allele is equivalent to the unmodified allele, that is it will produce the same phenotype and result in the transformation of substrate S, and such equivalent allele pairs are very common
- but if the allele produces a non-functional enzyme or no enzyme, the phenotype may be affected, and the trait will then depend only on the functioning of the unmodified allele
- the unmodified functioning allele, which represents the original phenotype, is the dominant allele, and the modified allele is generally the recessive allele, so the recessive trait is seen because of a non-functional enzyme or because no enzyme is produced

Co-dominance

- in co-dominance the F1 generation resembles both parents, rather than either of them as in dominance or being in between them as in incomplete dominance
- a good example is the different types of red blood cell that determine ABO blood grouping in human beings, controlled by the gene I, the plasma membrane of the red blood cells carrying sugar polymers that protrude from its surface, the kind of sugar being controlled by the gene
- the gene I has three alleles, I^A, I^B and i, the alleles I^A and I^B producing a slightly different form of the sugar while allele i produces no sugar, and because humans are diploid each person possesses any two of the three alleles
- I^A and I^B are completely dominant over i, so when I^A and i are present only I^A expresses and when I^B and i are present I^B expresses, but when I^A and I^B are present together they both express their own types of sugar, which is co-dominance, so the red blood cells have both A and B types of sugars

Table 4.B Table Showing the Genetic Basis of Blood Groups in Human Population

Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood types of offspring
I^A	I^A	I^AI^A	A
I^A	I^B	I^AI^B	AB
I^A	i	I^Ai	A
I^B	I^A	I^AI^B	AB
I^B	I^B	I^BI^B	B
I^B	i	I^Bi	B
i	i	ii	O

4.10 Multiple alleles

More than two alleles

- the example of ABO blood grouping also provides a good example of multiple alleles, since there are more than two, that is three, alleles governing the same character
- since in an individual only two alleles can be present, multiple alleles can be found only when population studies are made

One gene, more than one effect

- occasionally a single gene product may produce more than one effect, as with starch synthesis in pea seeds, controlled by one gene with two alleles B and b
- starch is synthesised effectively by BB homozygotes, which therefore produce large starch grains, while bb homozygotes have lesser efficiency and produce smaller grains, so after maturation BB seeds are round and bb seeds are wrinkled
- heterozygotes produce round seeds, so B seems to be the dominant allele, but the starch grains produced are of intermediate size in Bb seeds, so if starch grain size is considered as the phenotype the alleles show incomplete dominance
- dominance is therefore not an autonomous feature of a gene or of the product it has information for, but depends as much on the gene product and the production of a particular phenotype from it as on the particular phenotype we choose to examine

4.11 Inheritance of two genes

The dihybrid cross

- Mendel also crossed pea plants that differed in two characters, as in the cross between a pea plant with seeds of yellow colour and round shape and one with seeds of green colour and wrinkled shape
- he found that the seeds resulting from the crossing of the parents had yellow coloured and round shaped seeds, so yellow colour was dominant over green and round shape dominant over wrinkled, results identical to those from separate monohybrid crosses
- using Y for dominant yellow seed colour and y for recessive green, R for round shaped seeds and r for wrinkled, the genotype of the parents can be written as RRYY and rryy, and the gametes RY and ry unite on fertilisation to produce the F1 hybrid RrYy
- when Mendel self hybridised the F1 plants he found that 3/4th of F2 plants had yellow seeds and 1/4th green, and round and wrinkled seed shape also segregated in a 3:1 ratio, just as in a monohybrid cross

Parents RRYY, round and yellow, crossed with rryy, wrinkled and green

F1 all RrYy, round and yellow

Four gamete types from the F1, RY, Ry, rY and ry

Sixteen squares of the Punnett square filled from them

F2 phenotypes in the ratio 9 : 3 : 3 : 1

Figure 4.G Results of a dihybrid cross where the two parents differed in two pairs of contrasting traits: seed colour and seed shape

4.12 Law of independent assortment

The 9:3:3:1 ratio

- in the dihybrid cross the phenotypes round and yellow, wrinkled and yellow, round and green, and wrinkled and green appeared in the ratio 9:3:3:1, a ratio observed for several pairs of characters that Mendel studied
- the ratio can be derived as a combination series of 3 yellow : 1 green with 3 round : 1 wrinkled, written as (3 Round : 1 Wrinkled) (3 Yellow : 1 Green) = 9 Round, Yellow : 3 Wrinkled, Yellow : 3 Round, Green : 1 Wrinkled, Green
- based on such observations on dihybrid crosses, that is crosses between plants differing in two traits, Mendel proposed a second set of generalisations called the law of independent assortment, which states that when two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters

Following it through the gametes

- the Punnett square can be used to understand the independent segregation of the two pairs of genes during meiosis and the production of eggs and pollen in the F1 RrYy plant
- considering the segregation of the pair R and r, fifty per cent of the gametes have the gene R and the other 50 per cent r, and besides each gamete having either R or r it should also have the allele Y or y, the segregation of 50 per cent R and 50 per cent r being independent from that of 50 per cent Y and 50 per cent y
- therefore 50 per cent of the r bearing gametes has Y and the other 50 per cent has y, and similarly for the R bearing gametes, so there are four genotypes of gamete, RY, Ry, rY and ry, each with a frequency of 25 per cent or 1/4th of the total
- writing the four types of egg and pollen on the two sides of a Punnett square makes it easy to derive the composition of the zygotes that give rise to the F2 plants, which leaves the number of different genotypes and phenotypes among the 16 squares, and the genotypic ratio at F2, to be worked out

4.13 Chromosomal theory of inheritance

Why Mendel went unrecognised

- Mendel published his work on inheritance of characters in 1865 but for several reasons it remained unrecognised till 1900
- communication was not easy in those days and his work could not be widely publicised

A cell with four chromosomes entering meiosis

Homologous pairs aligning at the metaphase plate

Chromosomes segregating to opposite poles

Germ cells formed, each with one of each pair

Figure 4.H Meiosis and germ cell formation in a cell with four chromosomes. Can you see how chromosomes segregate when germ cells are formed?

The other reasons

- his concept of genes, or factors in Mendel's words, as stable and discrete units that controlled the expression of traits, and of a pair of alleles which did not blend with each other, was not accepted by his contemporaries as an explanation for the apparently continuous variation seen in nature
- his approach of using mathematics to explain biological phenomena was totally new and unacceptable to many of the biologists of his time
- and though his work suggested that factors were discrete units, he could not provide any physical proof for their existence or say what they were made of

Rediscovery, and the chromosomes

- in 1900 three scientists, de Vries, Correns and von Tschermak, independently rediscovered Mendel's results on the inheritance of characters
- by this time, due to advancements in microscopy, scientists were able to carefully observe cell division, which led to the discovery of structures in the nucleus that appeared to double and divide just before each cell division, called chromosomes, that is coloured bodies, as they were visualised by staining
- by 1902 the chromosome movement during meiosis had been worked out, and Walter Sutton and Theodore Boveri noted that the behaviour of chromosomes was parallel to the behaviour of genes and used chromosome movement to explain Mendel's laws
- the important things to remember are that chromosomes as well as genes occur in pairs, and that the two alleles of a gene pair are located on homologous sites on homologous chromosomes

Two chromosome pairs aligning at the metaphase plate

Possibility I, orange and green segregating together

Possibility II, orange segregating with red

Each alignment independent of the other

Figure 4.I Independent assortment of chromosomes

4.14 Chromosomes beside genes

Two alignments

- during anaphase of meiosis I the two chromosome pairs can align at the metaphase plate independently of each other
- in the first possibility one long orange and short green chromosome, and one long yellow and short red chromosome, arrive at the same pole
- in the second possibility one long orange and short red chromosome, and one long yellow and short green chromosome, arrive at the same pole

Table 4.C **A Comparison between the Behaviour of Chromosomes and Genes**

A	B
Occur in pairs	Occur in pairs
Segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete	Segregate at gamete formation and only one of each pair is transmitted to a gamete
Independent pairs segregate independently of each other	One pair segregates independently of another pair

Which of these columns A or B represents the chromosome and which represents the gene, and how you decided, is worth working out.

Drosophila melanogaster, the tiny fruit fly

The male, smaller, with a darker abdomen tip

The female, larger

The sexes easily distinguishable under low power

Figure 4.J Drosophila melanogaster (a) Male (b) Female

4.15 Sutton, Boveri and Morgan

The theory, and its verification

- Sutton and Boveri argued that the pairing and separation of a pair of chromosomes would lead to the segregation of the pair of factors they carried
- Sutton united the knowledge of chromosomal segregation with Mendelian principles and called it the chromosomal theory of inheritance
- experimental verification of that theory by Thomas Hunt Morgan and his colleagues led to discovering the basis for the variation that sexual reproduction produced
- Morgan worked with the tiny fruit flies, Drosophila melanogaster, which were very suitable for such studies, since they could be grown on simple synthetic medium in the laboratory, complete their life cycle in about two weeks, produce a large number of progeny from a single mating, show a clear differentiation of the sexes and carry many types of hereditary variation visible with low power microscopes

4.16 Linkage and recombination

What Morgan found

- Morgan carried out several dihybrid crosses in Drosophila to study genes that were sex-linked, the crosses being similar to the dihybrid crosses carried out by Mendel in peas
- for example he hybridised yellow-bodied, white-eyed females to brown-bodied, red-eyed males and intercrossed their F1 progeny, and observed that the two genes did not segregate independently of each other and that the F2 ratio deviated very significantly from the 9:3:3:1 ratio expected when the two genes are independent
- Morgan and his group knew that the genes were located on the X chromosome, and saw quickly that when the two genes in a dihybrid cross were situated on the same chromosome the proportion of parental gene combinations was much higher than the non-parental type
- Morgan attributed this to the physical association or linkage of the two genes, and coined the term linkage to describe this physical association of genes on a chromosome and the term recombination to describe the generation of non-parental gene combinations

Cross A, between the genes y and w

Cross B, between the genes w and m

Dominant wild type alleles marked with a plus sign in superscript

The linkage between y and w stronger than that between w and m

Figure 4.K Linkage: Results of two dihybrid crosses conducted by Morgan. Cross A shows crossing between gene y and w; Cross B shows crossing between genes w and m. Here dominant wild type alleles are represented with (+) sign in superscript

4.17 Mapping the chromosome

Tight and loose linkage

- Morgan and his group also found that even when genes were grouped on the same chromosome, some genes were very tightly linked, showing very low recombination, while others were loosely linked, showing higher recombination
- he found that the genes white and yellow were very tightly linked and showed only 1.3 per cent recombination, while white and miniature wing showed 37.2 per cent recombination
- his student Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes, and mapped their position on the chromosome
- today genetic maps are extensively used as a starting point in the sequencing of whole genomes, as was done in the case of the Human Genome Sequencing Project

4.18 Polygenic inheritance

Traits spread across a gradient

- Mendel's studies mainly described traits that have distinct alternate forms, such as flower colour which is either purple or white, but many traits are not so distinct in their occurrence and are spread across a gradient, as with human height, where we do not just have tall or short people as two distinct alternatives but a whole range of possible heights
- such traits are generally controlled by three or more genes and are thus called polygenic traits, and besides the involvement of multiple genes polygenic inheritance also takes into account the influence of environment, human skin colour being another classic example
- in a polygenic trait the phenotype reflects the contribution of each allele, that is the effect of each allele is additive
- assuming that three genes A, B and C control skin colour in humans, with the dominant forms A, B and C responsible for dark skin colour and the recessive forms a, b and c for light skin colour, the genotype with all dominant alleles (AABBCC) will have the darkest skin and that with all recessive alleles (aabbcc) the lightest, while a genotype with three dominant and three recessive alleles will have an intermediate skin colour, so the number of each type of allele determines the darkness or lightness of the skin

4.19 Pleiotropy

One gene, many phenotypes

- there are instances where a single gene can exhibit multiple phenotypic expression, and such a gene is called a pleiotropic gene
- the underlying mechanism of pleiotropy in most cases is the effect of a gene on metabolic pathways which contribute towards different phenotypes
- an example is the disease phenylketonuria, which occurs in humans, caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase, a single gene mutation, which manifests itself through phenotypic expression characterised by mental retardation and a reduction in hair and skin pigmentation

In humans and in Drosophila the female is XX, homogametic

The male is XY, heterogametic

In many birds the female has dissimilar chromosomes ZW

And the male two similar ZZ chromosomes

Figure 4.L Determination of sex by chromosomal differences: (a,b) Both in humans and in Drosophila, the female has a pair of XX chromosomes (homogametic) and the male XY (heterogametic) composition; (c) In many birds, female has a pair of dissimilar chromosomes ZW and male two similar ZZ chromosomes

4.20 Sex determination

Henking and the X body

- the mechanism of sex determination has always been a puzzle before the geneticists, and the initial clue about the genetic or chromosomal mechanism can be traced back to experiments carried out in insects, the cytological observations made in a number of them leading to the development of the concept
- Henking in 1891 could trace a specific nuclear structure all through spermatogenesis in a few insects, and observed that 50 per cent of the sperm received this structure after spermatogenesis whereas the other 50 per cent did not, and he named it the X body but could not explain its significance
- further investigations by other scientists led to the conclusion that the X body of Henking was in fact a chromosome, which is why it was given the name X-chromosome

The XO type

- in a large number of insects the mechanism of sex determination is of the XO type, that is all eggs bear an additional X-chromosome besides the other chromosomes, the autosomes, while some of the sperms bear the X-chromosome and some do not
- eggs fertilised by sperm having an X-chromosome become females and those fertilised by sperms that do not have an X-chromosome become males, which leaves the question of whether the number of chromosomes in the male and female are equal
- due to the involvement of the X-chromosome in the determination of sex it was designated the sex chromosome, and the rest of the chromosomes were named autosomes, the grasshopper being an example of XO type in which the males have only one X-chromosome besides the autosomes whereas females have a pair

The XY type

- in a number of other insects and mammals including man, XY type of sex determination is seen, where both male and female have the same number of chromosomes
- among the males an X-chromosome is present but its counterpart is distinctly smaller and called the Y-chromosome, while females have a pair of X-chromosomes, and both males and females bear the same number of autosomes
- so the males have autosomes plus XY while females have autosomes plus XX, and in human beings and in Drosophila the males have one X and one Y chromosome whereas females have a pair of X-chromosomes besides autosomes

An unfertilised egg developing into a male drone

A fertilised egg developing into a female, queen or worker

Females diploid with 32 chromosomes, males haploid with 16

Males producing sperms by mitosis

Figure 4.M Sex determination in honey bee

4.21 Male and female heterogamety

The two patterns

- in both the XO and the XY type, males produce two kinds of gamete, either with or without an X-chromosome, or some with X and some with Y, a mechanism designated male heterogamety
- in birds a different mechanism is observed, in which the total number of chromosomes is the same in both sexes but two kinds of gamete in terms of sex chromosomes are produced by females, that is female heterogamety
- to distinguish it from the mechanism described earlier, the two sex chromosomes of a female bird are designated Z and W, the female having one Z and one W whereas the male has a pair of Z-chromosomes besides the autosomes

4.22 Sex determination in humans

Which gamete decides

- out of the 23 pairs of chromosomes present, 22 pairs are the same in both sexes, and these are the autosomes, while a pair of X-chromosomes is present in the female and an X and a Y chromosome are determinant of the male characteristic
- during spermatogenesis two kinds of gamete are produced, 50 per cent of the sperm carrying the X-chromosome and the rest the Y, besides the autosomes, while females produce only one kind of ovum, with an X-chromosome
- there is an equal probability of fertilisation by a sperm carrying either X or Y, so an ovum fertilised by an X-carrying sperm develops into a female (XX) and by a Y-carrying sperm into a male, so it is the genetic makeup of the sperm that determines the sex of the child
- it is also evident that in each pregnancy there is always a 50 per cent probability of either a male or a female child, and it is unfortunate that in our society women are blamed for giving birth to female children and have been ostracised and ill-treated because of this false notion

4.23 Sex determination in the honey bee

Haplodiploidy

- sex determination in the honey bee rests on the number of sets of chromosomes an individual receives, an offspring from the union of a sperm and an egg developing as a female, queen or worker, and an unfertilised egg as a male drone by parthenogenesis
- this means the males have half the number of chromosomes of a female, the females being diploid with 32 chromosomes and the males haploid with 16
- this haplodiploid system has special features, the males producing sperms by mitosis, having no father and so no sons, but having a grandfather and possibly grandsons
- how the sex-determination mechanism differs in the birds, and whether the sperm or the egg is responsible for the sex of the chicks, are worth working out

Square for a male, circle for a female

Shaded symbol for an affected individual

A horizontal line joining a mated pair

A vertical line dropping to their offspring

Figure 4.N Symbols used in the human pedigree analysis

4.24 Mutation

What mutation is

- mutation is a phenomenon which results in alteration of DNA sequences and consequently in changes in the genotype and the phenotype of an organism, and in addition to recombination it is another phenomenon that leads to variation in DNA
- one DNA helix runs continuously from one end to the other in each chromatid, in a highly supercoiled form, so loss by deletion or gain by insertion or duplication of a segment of DNA results in alteration in chromosomes, and since genes are located on chromosomes such alteration results in abnormalities or aberrations, chromosomal aberrations being commonly observed in cancer cells
- mutation also arises due to change in a single base pair of DNA, which is known as point mutation, a classical example being sickle cell anemia, while deletions and insertions of base pairs cause frame-shift mutations
- there are many chemical and physical factors that induce mutations, referred to as mutagens, and UV radiations can cause mutations in organisms, so UV is a mutagen

4.25 Pedigree analysis

Reading a family tree

- the idea that disorders are inherited has been prevailing in human society since long, based on the heritability of certain characteristic features in families, and after the rediscovery of Mendel's work the practice of analysing inheritance patterns of traits in human beings began
- since the controlled crosses that can be performed in pea plants or some other organisms are not possible in human beings, study of the family history about inheritance of a particular trait provides an alternative, and such an analysis of traits in several generations of a family is called pedigree analysis
- in pedigree analysis the inheritance of a particular trait is represented in the family tree over generations, and in human genetics it provides a strong tool, used to trace the inheritance of a specific trait, abnormality or disease

An autosomal dominant trait, as in myotonic dystrophy

Affected individuals in every generation

An autosomal recessive trait, as in sickle-cell anaemia

Affected individuals from two unaffected carriers

Figure 4.0 Representative pedigree analysis of (a) Autosomal dominant trait (for example: Myotonic dystrophy) (b) Autosomal recessive trait (for example: Sickle-cell anaemia)

Where disorders come from

- each and every feature in any organism is controlled by one or the other gene located on the DNA present in the chromosome, and DNA is the carrier of genetic information, transmitted from one generation to the other without any change or alteration
- however, changes or alterations do take place occasionally, and such an alteration or change in the genetic material is referred to as mutation
- a number of disorders in human beings have been found to be associated with the inheritance of changed or altered genes or chromosomes

4.26 Mendelian disorders

Single gene disorders

- genetic disorders may broadly be grouped into two categories, Mendelian disorders and chromosomal disorders, the Mendelian ones being mainly determined by alteration or mutation in a single gene
- these disorders are transmitted to the offspring on the same lines as the principles of inheritance, and the pattern can be traced in a family by pedigree analysis, the most common and prevalent ones being haemophilia, cystic fibrosis, sickle-cell anaemia, colour blindness, phenylketonuria and thalassemia
- such disorders may be dominant or recessive, and by pedigree analysis one can easily understand whether the trait in question is dominant or recessive, while the trait may also be linked to the sex chromosome as in the case of haemophilia, an X-linked recessive trait that shows transmission from carrier female to male progeny

Colour blindness

- it is a sex-linked recessive disorder due to a defect in either the red or the green cone of the eye, resulting in failure to discriminate between red and green colour
- this defect is due to mutation in certain genes present in the X chromosome
- it occurs in about 8 per cent of males and only about 0.4 per cent of females, because the genes that lead to red-green colour blindness are on the X chromosome, and males have only one X chromosome while females have two

Red blood cells of a normal individual, biconcave discs

Red blood cells of a patient, elongated and sickle like

Glutamic acid at the sixth position of the beta chain

Valine there instead, in the sickle-cell haemoglobin

Figure 4.P Micrograph of the red blood cells and the amino acid composition of the relevant portion of β-chain of haemoglobin: (a) From a normal individual; (b) From an individual with sickle-cell anaemia

Colour blindness in a family

- the son of a woman who carries the gene has a 50 per cent chance of being colour blind
- the mother is not herself colour blind because the gene is recessive, that is its effect is suppressed by her matching dominant normal gene
- a daughter will not normally be colour blind, unless her mother is a carrier and her father is colour blind

4.27 Haemophilia

A sex linked recessive disease

- this disease shows its transmission from an unaffected carrier female to some of the male progeny, and has been widely studied
- in it a single protein that is part of the cascade of proteins involved in the clotting of blood is affected, so in an affected individual a simple cut will result in non-stop bleeding
- the heterozygous carrier female may transmit the disease to sons, and the possibility of a female becoming haemophilic is extremely rare, because the mother of such a female has to be at least a carrier and the father should be haemophilic
- the family pedigree of Queen Victoria shows a number of haemophilic descendents, as she was a carrier of the disease

4.28 Sickle-cell anaemia

An autosome linked recessive trait

- it can be transmitted from parents to offspring when both the partners are carriers for the gene, that is heterozygous
- the disease is controlled by a single pair of alleles, HbA and HbS, and out of the three possible genotypes only individuals homozygous for HbS show the diseased phenotype
- heterozygous individuals appear apparently unaffected but are carriers, as there is a 50 per cent probability of transmission of the mutant gene to the progeny, thus exhibiting sickle-cell trait

What the sickle-cell defect is

- the defect is caused by the substitution of glutamic acid (Glu) by valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule
- the substitution of the amino acid in the globin protein results from a single base substitution at the sixth codon of the beta globin gene, from GAG to GUG
- the mutant haemoglobin molecule undergoes polymerisation under low oxygen tension, causing the change in the shape of the RBC from a biconcave disc to an elongated sickle like structure

Phenylketonuria

- this inborn error of metabolism is also inherited as an autosomal recessive trait
- the affected individual lacks an enzyme that converts the amino acid phenylalanine into tyrosine, so phenylalanine is accumulated and converted into phenylpyruvic acid and other derivatives
- accumulation of these in the brain results in mental retardation, and they are also excreted through urine because of poor absorption by the kidney

4.29 Thalassemia

A quantitative problem

- this is also an autosome-linked recessive blood disease, transmitted from parents to offspring when both the partners are unaffected carriers for the gene
- the defect could be due to either mutation or deletion, which ultimately results in a reduced rate of synthesis of one of the globin chains, α and β, that make up haemoglobin, causing the formation of abnormal haemoglobin molecules and so the anaemia characteristic of the disease
- it can be classified according to which chain is affected, so in α thalassemia production of the α globin chain is affected while in β thalassemia production of the β globin chain is affected
- α thalassemia is controlled by two closely linked genes HBA1 and HBA2 on chromosome 16 of each parent and is observed due to mutation or deletion of one or more of the four genes, the more genes affected the less alpha globin produced, while β thalassemia is controlled by a single gene HBB on chromosome 11 of each parent and occurs due to mutation of one or both genes, and thalassemia differs from sickle-cell anaemia in that the former is a quantitative problem of synthesising too few globin molecules while the latter is a qualitative problem of synthesising an incorrectly functioning globin

4.30 Chromosomal disorders

Aneuploidy and polyploidy

- chromosomal disorders are caused by absence or excess or abnormal arrangement of one or more chromosomes
- failure of segregation of chromatids during the cell division cycle results in the gain or loss of a chromosome, called aneuploidy, so Down's syndrome results in the gain of an extra copy of chromosome 21 and Turner's syndrome from loss of an X chromosome in human females
- failure of cytokinesis after the telophase stage of cell division results in an increase in a whole set of chromosomes, a phenomenon known as polyploidy, often seen in plants

Flat back of head and broad flat face

Big and wrinkled tongue, partially open mouth

Broad palm with a characteristic palm crease

Many loops on finger tips, and congenital heart disease

The karyotype carrying three copies of chromosome 21

Figure 4.Q A representative figure showing an individual inflicted with Down's syndrome and the corresponding chromosomes of the individual

Tall stature with feminised character, in Klinefelter Syndrome

Short stature and underdeveloped feminine character, in Turner's Syndrome

One from an extra X chromosome, the other from a missing one

Figure 4.R Diagrammatic representation of genetic disorders due to sex chromosome composition in humans : (a) Klinefelter Syndrome; (b) Turner's Syndrome

4.31 Trisomy and monosomy

The counts

- the total number of chromosomes in a normal human cell is 46, that is 23 pairs, of which 22 pairs are autosomes and one pair are sex chromosomes
- sometimes, though rarely, either an additional copy of a chromosome may be included in an individual or an individual may lack one of a pair, situations known as trisomy or monosomy of a chromosome respectively
- such a situation leads to very serious consequences in the individual, and Down's syndrome, Turner's syndrome and Klinefelter's syndrome are common examples of chromosomal disorders

The three syndromes

- the cause of Down's syndrome is the presence of an additional copy of chromosome number 21, that is trisomy of 21, a disorder first described by Langdon Down in 1866, the affected individual being short statured with a small round head, furrowed tongue and partially open mouth, a broad palm with a characteristic palm crease, and retarded physical, psychomotor and mental development
- Klinefelter's syndrome is caused by the presence of an additional copy of the X-chromosome, resulting in a karyotype of 47, XXY, such an individual having overall masculine development but also expressing feminine development, that is development of breast or gynaecomastia, and being sterile
- Turner's syndrome is caused by the absence of one of the X chromosomes, that is 45 with X0, such females being sterile as the ovaries are rudimentary, besides other features including lack of other secondary sexual characters

4.32 Summary

Mendel's laws

- genetics is a branch of biology which deals with the principles of inheritance and its practices, and Mendel was the first to study systematically the resemblance of progeny to their parents in morphological and physiological features
- while studying the pattern of inheritance in pea plants of contrasting characters he proposed the principles of inheritance, today referred to as Mendel's laws of inheritance, and proposed that the factors, later named genes, regulating the characters are found in pairs known as alleles
- he observed that the expression of the characters in the offspring follows a definite pattern in the first generation (F1), the second (F2) and so on, some characters being dominant over others, the dominant characters being expressed when factors are in heterozygous condition, which is the law of dominance, while recessive characters are only expressed in homozygous conditions
- the characters never blend in heterozygous condition, and a recessive character not expressed in heterozygous condition may be expressed again when it becomes homozygous, so characters segregate while gametes are formed, which is the law of segregation

Beyond simple dominance

- not all characters show true dominance, some showing incomplete dominance and some co-dominance
- when Mendel studied the inheritance of two characters together it was found that the factors independently assort and combine in all permutations and combinations, which is the law of independent assortment, and different combinations of gametes are theoretically represented in a square tabular form known as the Punnett square
- the factors, now known as genes, on chromosomes regulating the characters are called the genotype, and the physical expression of the chraracters is called phenotype
- after knowing that the genes are located on the chromosomes, a good correlation was drawn between Mendel's laws of segregation and assortment and the behaviour of chromosomes during meiosis, and the laws were extended in the form of the chromosomal theory of inheritance

Linkage, sex and mutation

- later it was found that the law of independent assortment does not hold true for genes located on the same chromosome, such genes being called linked genes, closely located genes assorting together and distantly located ones assorting independently due to recombination, so linkage maps corresponded to the arrangement of genes on a chromosome
- many genes were linked to sexes also, and called sex-linked genes, the two sexes having a set of chromosomes which were common and another set which was different, the different ones named sex chromosomes and the remaining set autosomes, so in humans a normal female has 22 pairs of autosomes and a pair of sex chromosomes XX and a male 22 pairs of autosomes and a pair XY, while in chicken sex chromosomes in the male are ZZ and in females ZW
- mutation is defined as change in the genetic material, a point mutation being a change of a single base pair in DNA, and sickle-cell anemia is caused by change of one base in the gene coding for the beta-chain of hemoglobin, while inheritable mutations can be studied by generating a pedigree of a family

4.33 Summary, continued

Whole chromosome changes

- some mutations involve changes in a whole set of chromosomes, that is polyploidy, or change in a subset of chromosome number, that is aneuploidy, which helped in understanding the mutational basis of genetic disorders
- Down's syndrome is due to trisomy of chromosome 21, where there is an extra copy of chromosome 21 and consequently the total number of chromosomes becomes 47
- in Turner's syndrome one X chromosome is missing and the sex chromosome is XO, and in Klinefelter's syndrome the condition is XXY, and these can be easily studied by analysis of karyotypes

4.34 Exercises

From this chapter

1. Mention the advantages of selecting pea plant for experiment by Mendel.
2. Differentiate between the following – (a) Dominance and Recessive (b) Homozygous and Heterozygous (c) Monohybrid and Dihybrid.
3. A diploid organism is heterozygous for 4 loci, how many types of gametes can be produced?
4. Explain the Law of Dominance using a monohybrid cross.
5. Define and design a test-cross.
6. Using a Punnett Square, workout the distribution of phenotypic features in the first filial generation after a cross between a homozygous female and a heterozygous male for a single locus.
7. When a cross in made between tall plant with yellow seeds (TtYy) and tall plant with green seed (Ttyy), what proportions of phenotype in the offspring could be expected to be (a) tall and green. (b) dwarf and green.
8. Two heterozygous parents are crossed. If the two loci are linked what would be the distribution of phenotypic features in F1 generation for a dibybrid cross?
9. Briefly mention the contribution of T.H. Morgan in genetics.
10. What is pedigree analysis? Suggest how such an analysis, can be useful.
11. How is sex determined in human beings?
12. A child has blood group O. If the father has blood group A and mother blood group B, work out the genotypes of the parents and the possible genotypes of the other offsprings.
13. Explain the following terms with example (a) Co-dominance (b) Incomplete dominance
14. What is point mutation? Give one example.
15. Who had proposed the chromosomal theory of the inheritance?
16. Mention any two autosomal genetic disorders with their symptoms.